

ANALYSIS OF:  
“ELECTIONS, PROTEST AND  
TRUST IN GOVERNMENT: A  
NATURAL EXPERIMENT FROM  
RUSSIA”  
AN AI REPLICATION

Kassandra Gervasi,  
Borna Atrchian,  
Bri Anderson

# A LITTLE RECAP: THE PAPER

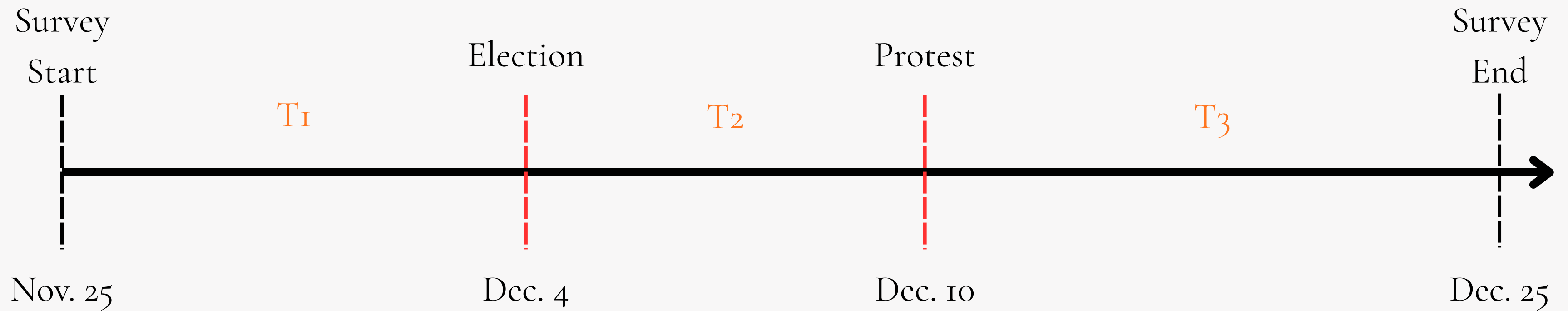
## Elections, Protest, and Trust in Government: A Natural Experiment from Russia

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**Timothy Frye**, Columbia University and National Research University Higher School of Economics  
**Ekaterina Borisova**, National Research University Higher School of Economics

How do elections and postelection protest shape political trust in a competitive autocracy? Taking advantage of largely exogenous variation in the timing of a survey conducted in Moscow in 2011, we find that with few exceptions the election had little systematic effect on political trust, perhaps because vote improprieties were not new information. In contrast, the unexpected protest that followed increased trust in government. In this case, heightened trust arises largely from opposition voters—those most likely to be surprised by permission to hold the protest—who update their beliefs. We argue that when autocrats permit protest unexpectedly, citizens may update their beliefs about the trustworthiness of the government. Our results suggest that citizens may cue not off the content of a protest but off the government's decision to permit it. In addition, autocrats can increase trust in government by allowing protest when it is unexpected.

# TIMELINE



# METHODS



We decided to use 2 different methods to conduct the AI replication analysis:

I. Simple AI Replication

II. AI Directed Replication

And in-between the two, we cleared Claude's memory and data.

**Why?**

Interested in seeing the potential differences between a strict AI report and one with some added direction.



# I. SIMPLE AI REPLICATION



# The Prompts

## PROMPT 1.1:

Hi Claude,

We want to replicate a published political science paper that investigates how elections and post-election protest shape political trust in a competitive autocracy. We are uploading the data as a .dta file entitled “moscowpopsurvey.dta”. We are also including the original paper’s code, which includes three other documents in .do format. Some of the data may be in Russian which will need to be translated to English. Using all of these data and files, please conduct a replication and produce a final report with all relevant findings and possible robustness checks.

## PROMPT 1.2:

We will now upload the original paper into the “final proj” folder, as a PDF entitled “replication\_paper.pdf”. Then produce a report comparing your findings to the original paper.

# II. AI DIRECTED REPLICATION



# The Prompt

## PROMPT 2.

We want to replicate a published political science paper that investigates how elections and postelection protest shape political trust in a competitive autocracy. We will also attach the paper as a PDF entitled “replication\_paper.pdf”. We are also uploading the data as a .dta file entitled “moscowpopsurvey.dta”. The original paper’s data also includes three other documents in .do format, which include the authors’ code. Please use all of this information to achieve the following goals:

The goals we want you to do are:

- Take and analyse/read all of the provided files
- Translate everything to English, as some of the original data is in Russian
- Translate the Stata code to R format,
- Replicate the paper’s findings using R format,
- Compare our replicated results with the original paper's results.
- Conduct the following checks:
- Power analysis (especially for sub groups) add minimum detectable effect calculations here, sensitivity analysis for power across different levels of alpha (since if a Bonferroni correction is applied the effective alpha drops to .005).

- Code integrity checks
- Unit testing
- Check the methodological choices the authors made (decisions about measurement + data inclusion/exclusion. Verify if anything is weird? Could anything have been improved?
- Verify the Estimators/Estimands/Quantities of Interest (e.g. maybe they're actually estimating ATT instead of ATE, etc. Or do they have the right QoI?, right error correction + SEs?).
- Anything else missing that would be of interest in a data analysis / quantitative analysis / causal inference class.
- Is this actually exogenous? Test this.
- Unequal groups and group composition may be an issue? Do you agree? If so, how and why? (men, women, ideology etc and other controls).
- Do balance checks for ALL variables. Might be better to ask explicitly for an omnibus test with SMDs included.



# The Prompt (cont'd)

- Test if period 2 (Dec. 5-9), which is the period that allows authors to separate election effect from protest effect, might be contaminated by anticipation of the December 10 protest which was announced December 7. Respondents interviewed on Dec. 8 and 9 may have known a protest was imminent, test whether results are sensitive to redefining period 2 as just December 5 & 6 (pre protest announcement) vs. December 7 to 9 (after announcement). If the period 2 post election estimates change, it suggests the presence of anticipation effects.

Using all of these data and files, please conduct a replication and produce a final report with all relevant findings and comparisons.

```

Welcome back Kassandra!

Sonnet 4.6 · Claude Pro ·
kassandragervasi@hotmail.com's Organization
/Users/kassandragervasi

Run /init to create a CLAUDE.md file with instructions for Claude
Note: You have launched claude in your home directory. For the best experience, launch it in a p...

Recent activity
No recent activity

* Voice mode is now available · /voice to enable

> /clear
└ (no content)

> Hi Claude,
All files are located in the folder: /Users/kassandragervasi/Documents/CLAUDE'S FOLDER/666/Final
Proj/.
We want to replicate a published political science paper that investigates how elections and postelection protest shape political trust in a competitive autocracy. We will also attach the paper as a PDF entitled
"replication.paper.pdf". We are also uploading the data as a .dta file entitled "moscowpopsurvey.dta". The original paper's data also includes three other documents in .do format, which include the authors' code.
Please use all of this information to achieve the following goals:
The goals we want you to do are:
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Conduct the following checks:
Power analysis (especially for sub groups) add minimum detectable effect calculations here, sensitivity analysis for power across different levels of alpha (since if a Bonferroni correction is applied the
effective alpha drops to .005).
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Check the methodological choices the authors made (decisions about measurement + data inclusion/exclusion. Verify if anything is weird? Could anything have been improved?
Verify the Estimators/Estimands/Quantities of Interest (e.g. maybe they're actually estimating ATT instead of ATE, etc. Or do they have the right QoI?, right error correction + SEs?).
Anything else missing that would be of interest in a data analysis / quantitative analysis / causal inference class.
Is this actually as-if random? This is a natural experiment that resides on this, but does this data satisfy these conditions 'as-if' random'?
Is this actually exogenous? Test this.
Unequal groups and group composition may be an issue? Do you agree? If so, how and why? (men, women, ideology etc and other controls).
Do balance checks for ALL variables. Might be better to ask explicitly for an omnibus test with SMDs included.
Test if period 2 (Dec. 5-9), which is the period that allows authors to separate election effect from protest effect, might be contaminated by anticipation of the December 10 protest which was announced December
7. Respondents interviewed on Dec. 8 and 9 may have known a protest was imminent, test whether results are sensitive to redefining period 2 as just December 5 & 6 (pre protest announcement) vs. December 7 to 9
(after announcement). If the period 2 post election estimates change, it suggests the presence of anticipation effects.
McCrory Density Test at cutoff dates to test as-if random assumption.
Apply Bonferroni and Benjamini-Hochberg corrections across all institutional models represented in Tables 2 and 3. Likely that some of the institutional trust models/findings won't survive these corrections since
they're at the margin of statistical significance.

Using all of these data and files, please conduct a replication and produce a final report with all relevant findings and comparisons.
```

# RESULTS

# I. SIMPLE AI REPLICATION

# Simple AI Replication Findings

- Findings were largely replicated
- Biggest problem: Claude swapped pre-election and post-election data
  - Concluded that the trends were reversed (though with the same significance)
- Most common discrepancies were for measure of Trust in Government, particularly on Protest Day
- Some errors in response rates
- However, additional robustness checks were all completed successfully (additional demographic controls, F-tests, ordered logit)



# Simple AI Replication Findings





# Simple AI Replication Findings

Paper

Table 1. Trust in Institutions

|                                | Period 1<br>Preelection<br>before 12/4 | Period 2<br>Postelection<br>12/5–12/9 | Period 3<br>Postprotest<br>after 12/10 |
|--------------------------------|----------------------------------------|---------------------------------------|----------------------------------------|
| Trust in government index      | 2.67<br>(.77)                          | 2.75<br>(.83)                         | 2.92 <sup>b,c</sup><br>(.76)           |
| Army                           | 2.90<br>(1.06)                         | 3.15 <sup>a</sup><br>(1.03)           | 3.18 <sup>c</sup><br>(1.00)            |
| Police                         | 2.48<br>(.87)                          | 2.77 <sup>a</sup><br>(.96)            | 2.86 <sup>c</sup><br>(.97)             |
| FSB (Federal Security Service) | 2.68<br>(1.03)                         | 2.96 <sup>a</sup><br>(1.04)           | 3.06 <sup>c</sup><br>(.97)             |
| Courts                         | 2.55<br>(.94)                          | 2.68<br>(1.00)                        | 2.76 <sup>c</sup><br>(.95)             |
| Municipal government           | 2.68<br>(.95)                          | 2.70<br>(.98)                         | 2.91 <sup>b,c</sup><br>(.97)           |
| Federal government             | 2.75<br>(1.02)                         | 2.72<br>(1.02)                        | 2.98 <sup>b,c</sup><br>(.98)           |
| Duma (parliament)              | 2.53<br>(1.01)                         | 2.50<br>(.94)                         | 2.66 <sup>b</sup><br>(.99)             |
| Procuracy                      | 2.77<br>(.94)                          | 2.78<br>(.99)                         | 2.92 <sup>b,c</sup><br>(.98)           |
| United Nations                 | 3.07<br>(1.01)                         | 2.95<br>(1.04)                        | 3.04<br>(1.08)                         |

Table 3: Table 1A: Pre-Election vs. Post-Election Period (Period 1 vs. Period 2)

|             | Mean Pre-Elec | Mean Post-Elec | Difference | t-stat | p-value |
|-------------|---------------|----------------|------------|--------|---------|
| Army        | 3.147         | 2.898          | -0.249     | 2.796  | 0.005   |
| Police      | 2.771         | 2.480          | -0.291     | 3.753  | 0.000   |
| FSB         | 2.962         | 2.681          | -0.281     | 3.103  | 0.002   |
| Court       | 2.677         | 2.550          | -0.127     | 1.525  | 0.128   |
| City Gov't  | 2.695         | 2.680          | -0.015     | 0.178  | 0.859   |
| Fed Gov't   | 2.720         | 2.749          | 0.029      | -0.330 | 0.742   |
| Duma        | 2.498         | 2.529          | 0.031      | -0.363 | 0.717   |
| Procuracy   | 2.776         | 2.774          | -0.002     | 0.023  | 0.982   |
| UN          | 2.949         | 3.073          | 0.124      | -1.318 | 0.188   |
| Gov't Index | 2.787         | 2.671          | -0.116     | 1.752  | 0.080   |

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# Simple AI Replication Findings

Paper

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| Gov't Index | 2.787         | 2.671          | -0.116     | 1.752  | 0.080   |

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# Simple AI Replication Findings

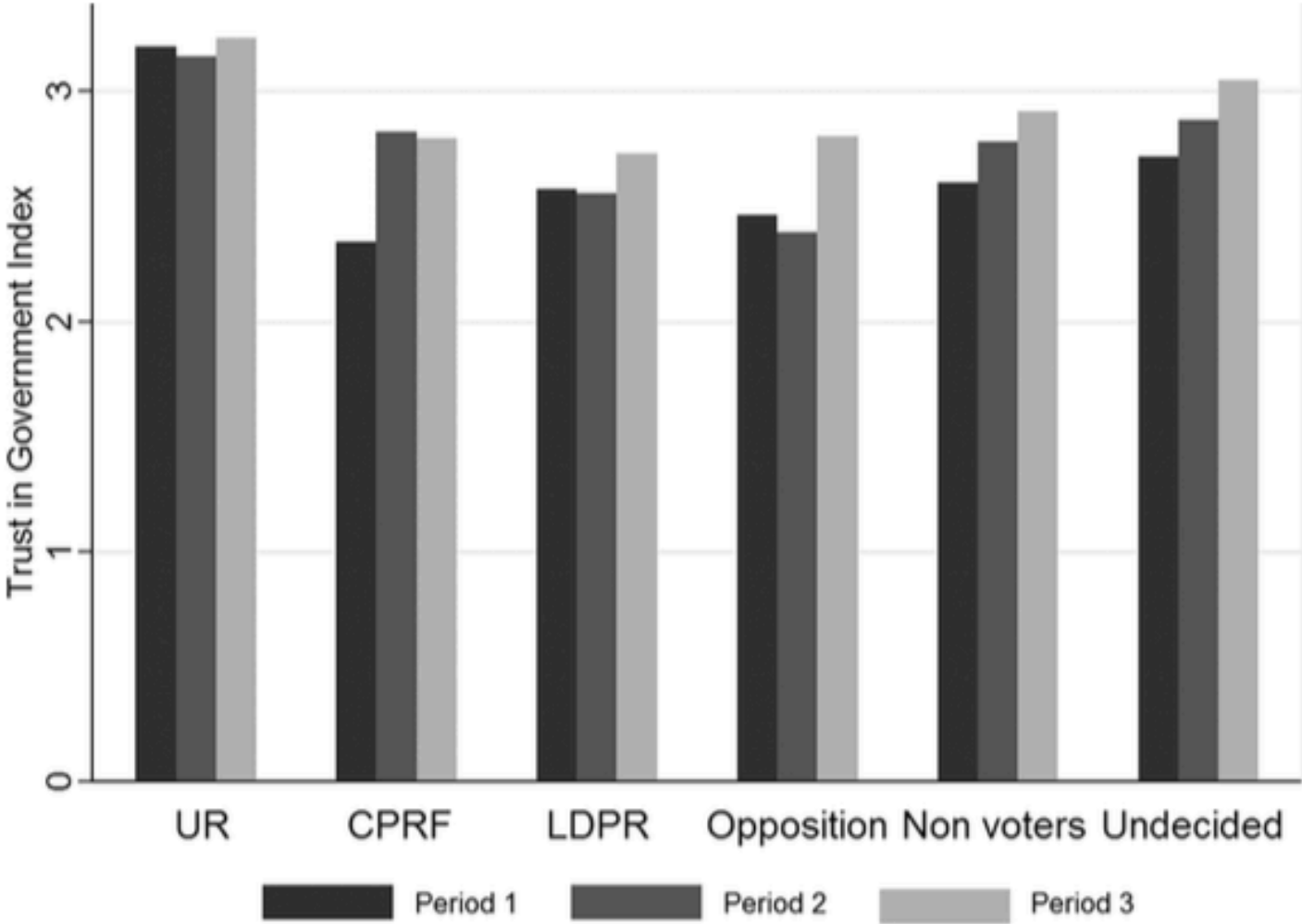


Figure 1. Partisanship and trust in government

*Paper*

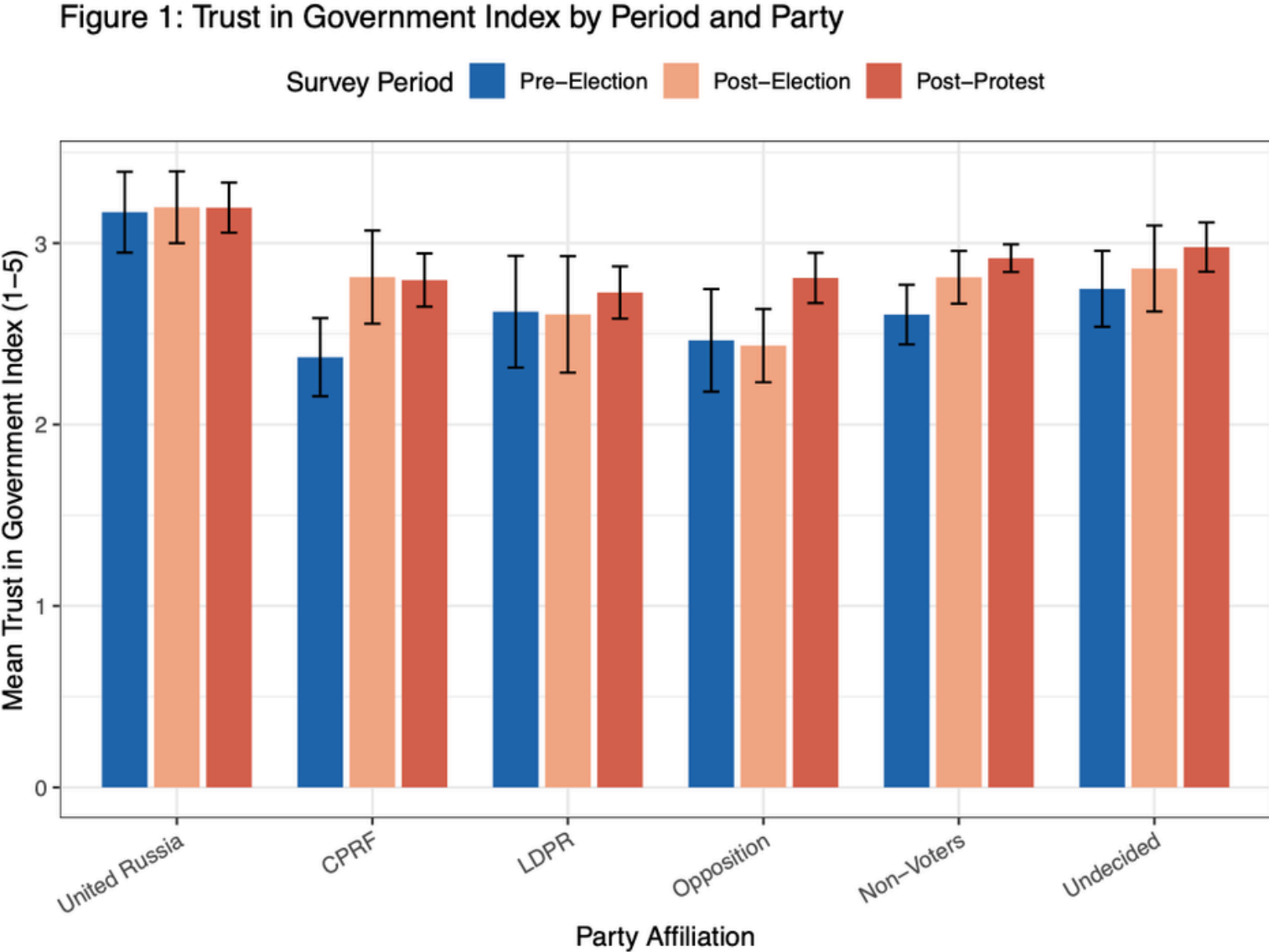


Figure 1: Mean Trust in Government Index across Three Survey Periods by Party Affiliation

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# Simple AI Replication Findings

*Paper*

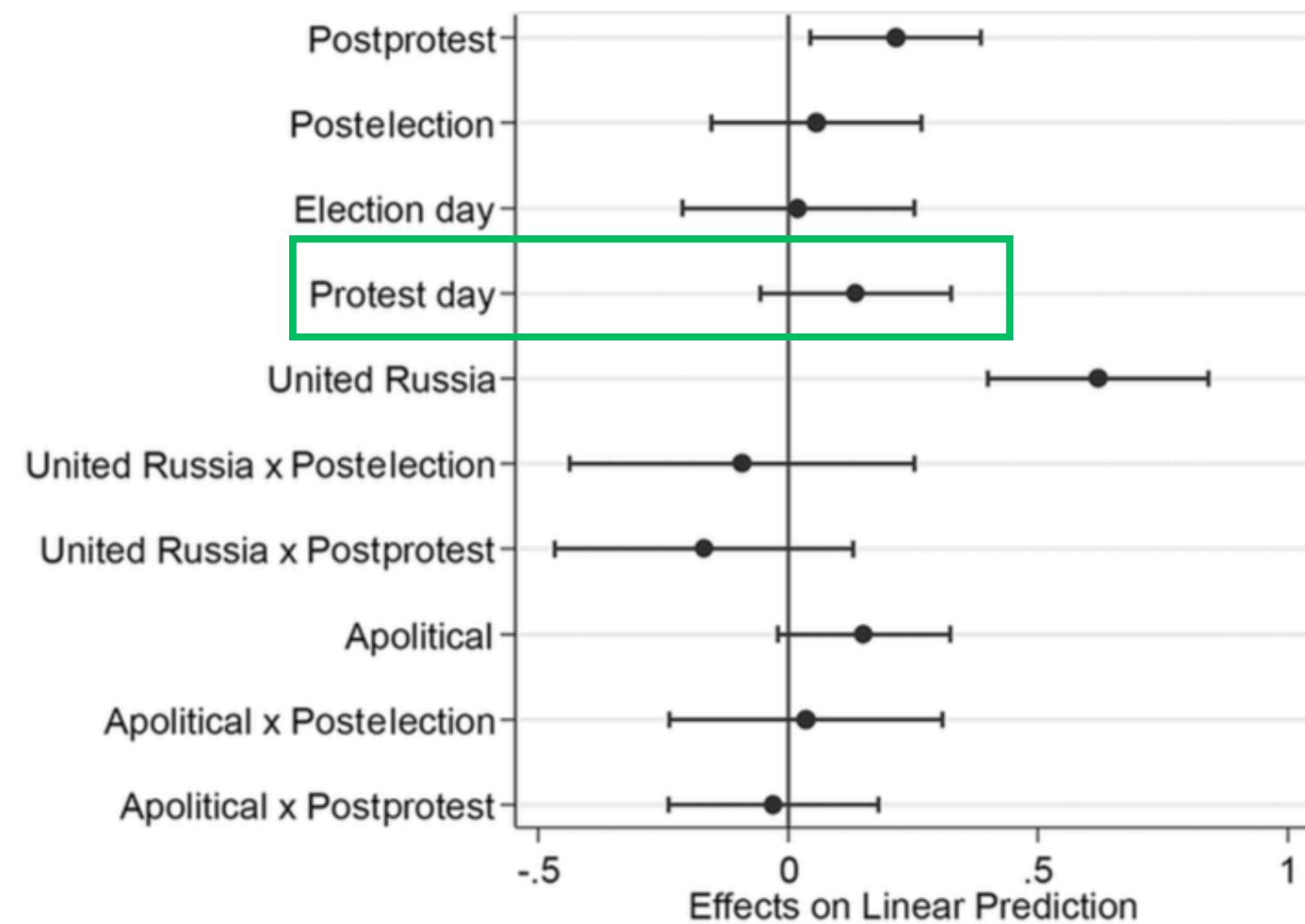
|              | Government |       |        |       |       | City       | Federal    |       |           |       |
|--------------|------------|-------|--------|-------|-------|------------|------------|-------|-----------|-------|
|              | Index      | Army  | Police | FSB   | Court | Government | Government | Duma  | Procuracy | UN    |
|              | (1)        | (2)   | (3)    | (4)   | (5)   | (6)        | (7)        | (8)   | (9)       | (10)  |
| Postprotest  | .16*       | .23*  | .28*   | .27*  | .13   | .17*       | .17*       | .05   | -.01      | .05   |
|              | (.06)      | (.08) | (.07)  | (.08) | (.07) | (.07)      | (.08)      | (.08) | (.08)     | (.09) |
| Postelection | .04        | .28*  | .24*   | .18*  | .08   | -.04       | -.05       | -.06  | -.14      | -.11  |
|              | (.07)      | (.09) | (.08)  | (.09) | (.09) | (.08)      | (.09)      | (.09) | (.09)     | (.10) |
| Protest day  | .13        | .50*  | .38*   | .28*  | .20   | .14        | .07        | -.09  | -.12      | .05   |
|              | (.10)      | (.12) | (.11)  | (.12) | (.11) | (.12)      | (.12)      | (.12) | (.12)     | (.13) |
| Election day | -.03       | -.09  | .22    | .21   | .11   | -.21       | -.28       | -.21  | -.04      | -.49* |
|              |            |       |        |       |       |            |            | (.14) | (.15)     | (.17) |

| Variable       | Army   | Police  | FSB     | Court  | City Gov't | Fed Gov't | Duma   | Procuracy | UN     | Gov't Index |
|----------------|--------|---------|---------|--------|------------|-----------|--------|-----------|--------|-------------|
| Post-Protest   | 0.23*  | 0.28*** | 0.27*** | 0.13   | 0.17*      | 0.17*     | 0.05   | -0.01     | 0.05   | 0.16*       |
|                | (0.08) | (0.07)  | (0.08)  | (0.07) | (0.07)     | (0.08)    | (0.08) | (0.08)    | (0.09) | (0.06)      |
| Post-Election  | 0.28*  | 0.24*   | 0.18*   | 0.08   | -0.04      | -0.05     | -0.06  | -0.14     | -0.11  | 0.06        |
|                | (0.09) | (0.08)  | (0.09)  | (0.09) | (0.08)     | (0.09)    | (0.09) | (0.09)    | (0.1)  | (0.07)      |
| Protest Day    | 0.5*** | 0.38*** | 0.28*   | 0.2    | 0.14       | 0.07      | -0.09  | -0.12     | 0.05   | 0.18*       |
|                | (0.12) | (0.11)  | (0.12)  | (0.11) | (0.12)     | (0.12)    | (0.12) | (0.12)    | (0.13) | (0.08)      |
| Election Day   | -0.09  | 0.22    | 0.21    | 0.11   | -0.21      | -0.28     | -0.21  | -0.04     | -      | -0.04       |
|                | (0.16) | (0.16)  | (0.17)  | (0.15) | (0.15)     | (0.15)    | (0.14) | (0.15)    | (0.17) | (0.12)      |
| R <sup>2</sup> | 0.071  | 0.056   | 0.088   | 0.056  | 0.072      | 0.085     | 0.062  | 0.054     | 0.042  | 0.079       |

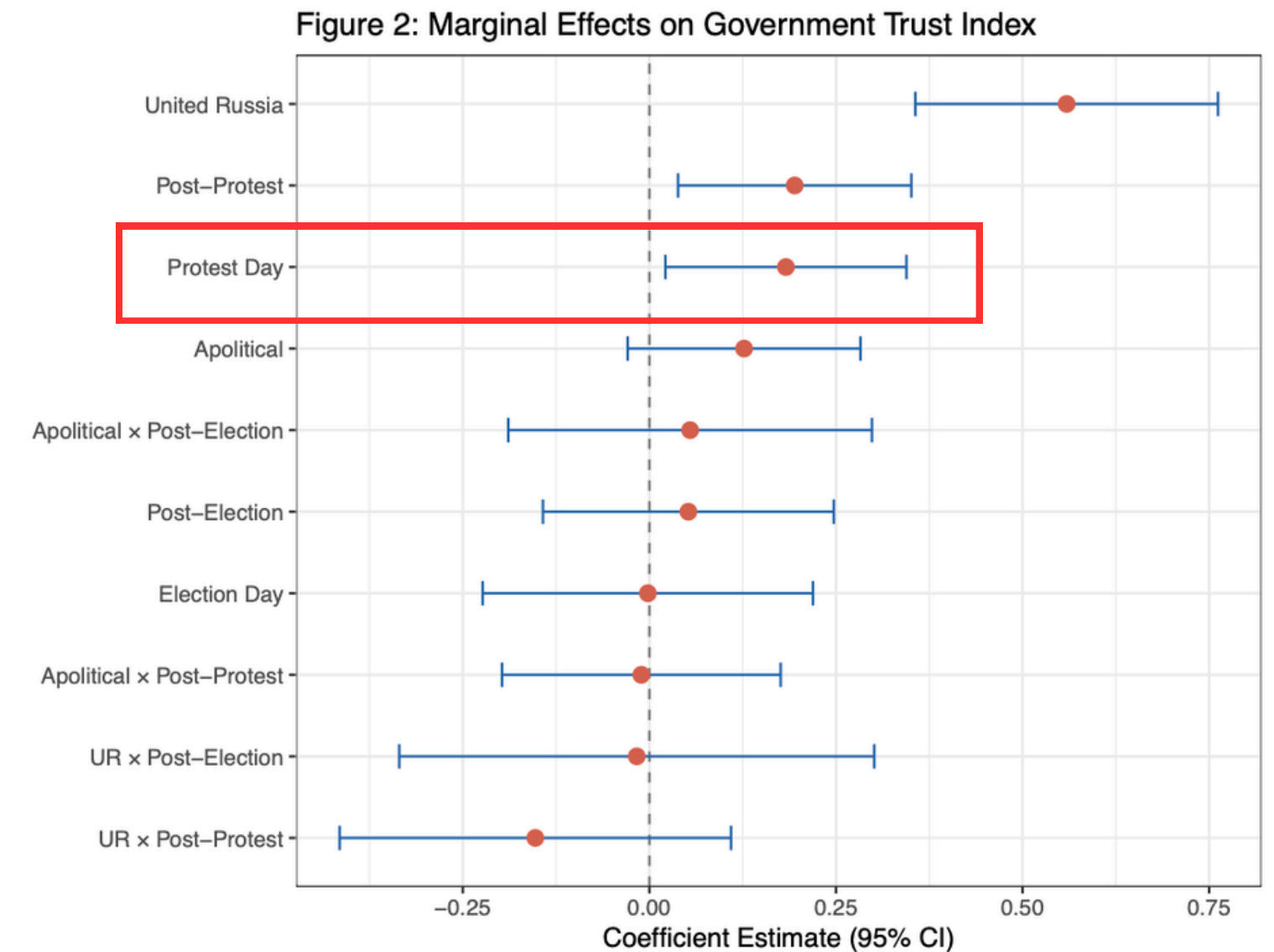
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(Some Robust SEs are also slightly different)

# Simple AI Replication Findings



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# Simple AI Replication Findings

Appendix II. Balance tests

|                              | Period 1<br>Pre-Election<br>Before 12/4 | Period2<br>Post-Election<br>12/5-12/9 | Period3<br>Post-Protest<br>After 12/10 |
|------------------------------|-----------------------------------------|---------------------------------------|----------------------------------------|
| Age                          | 42<br>(15)                              | 44<br>(15)                            | 44<br>(16)                             |
| Male                         | .43<br>(.50)                            | .46<br>(.50)                          | .48<br>(.50)                           |
| Education                    | 5.55<br>(1.78)                          | 5.37<br>(1.71)                        | 5.11 <sup>b c</sup><br>(1.69)          |
| Wealth                       | 3.73<br>(.67)                           | 3.80<br>(.70)                         | 3.62 <sup>b c</sup><br>(.72)           |
| Permanent Resident           | .94<br>(.25)                            | .96<br>(.20)                          | .96<br>(.21)                           |
| Russian ethnicity            | .91<br>(.29)                            | .86<br>(.35)                          | .90<br>(.30) <sup>b</sup>              |
| Not working                  | .07<br>(.25)                            | .05<br>(.22)                          | .08<br>(.27)                           |
| State Sector Work            | .02<br>(.13)                            | .04<br>(.20)                          | .02<br>(.16)                           |
| UR supporter                 | .17<br>(.38)                            | .14<br>(.34)                          | .14<br>(.34)                           |
| Communist Party<br>Supporter | .16<br>(.37)                            | .12<br>(.32)                          | .12<br>(.33)                           |
| Opposition Supporter         | .13<br>(.34)                            | .16<br>(.37)                          | .14<br>(.35)                           |

*Paper*

Table 11: Appendix II-A: Covariate Balance, Pre-Election vs. Post-Election

|                     | Mean Pre-Elec | Mean Post-Elec | p-value |
|---------------------|---------------|----------------|---------|
| Age                 | 43.513        | 42.239         | 0.324   |
| Male                | 0.459         | 0.433          | 0.535   |
| Education           | 5.369         | 5.551          | 0.220   |
| Wealth              | 3.797         | 3.728          | 0.244   |
| Permanent Resident  | 0.959         | 0.935          | 0.209   |
| Russian Nationality | 0.859         | 0.907          | 0.078   |
| Unemployed          | 0.053         | 0.069          | 0.443   |

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# Simple AI Replication Findings

Appendix IV. Non-responses

|                                 | Whole<br>sample | Period 1<br>Pre-<br>Election<br>Before 12/4 | Period2<br>Post-<br>Election<br>12/5-12/9 | Period3<br>Post-<br>Protest<br>After 12/10 |
|---------------------------------|-----------------|---------------------------------------------|-------------------------------------------|--------------------------------------------|
| Army                            | .02<br>(.15)    | .01<br>(.09)                                | .03<br>(.16)                              | .02<br>(.16)                               |
| Police                          | .01<br>(.12)    | .01<br>(.06)                                | .02<br>(.14)                              | .01<br>(.11)                               |
| FSB (Federal Security<br>Organ) | .12<br>(.33)    | .04<br>(.19)                                | .10 <sup>a</sup><br>(.30)                 | .15 <sup>b c</sup><br>(.36)                |
| Courts                          | .05<br>(.22)    | .02<br>(.14)                                | .05 <sup>a</sup><br>(.22)                 | .05 <sup>c</sup><br>(.23)                  |
| Municipal<br>Government         | .03<br>(.16)    | .01<br>(.11)                                | .05 <sup>a</sup><br>(.21)                 | .02 <sup>b</sup><br>(.14)                  |
| Federal Government              | .03<br>(.16)    | .02<br>(.13)                                | .03<br>(.17)                              | .03<br>(.16)                               |
| Duma (Parliament)               | .04<br>(.19)    | .02<br>(.14)                                | .05<br>(.21)                              | .04<br>(.19)                               |
| Procuracy                       | .06<br>(.24)    | .02<br>(.13)                                | .05 <sup>a</sup><br>(.22)                 | .07 <sup>c</sup><br>(.26)                  |
| United Nations                  | .21<br>(.40)    | .11<br>(.32)                                | .20 <sup>a</sup><br>(.40)                 | .24 <sup>c</sup><br>(.43)                  |
| Trust in Government<br>Index    | .19<br>(.40)    | .08<br>(.27)                                | .18 <sup>a</sup><br>(.39)                 | .22 <sup>c</sup><br>(.42)                  |

*Paper*

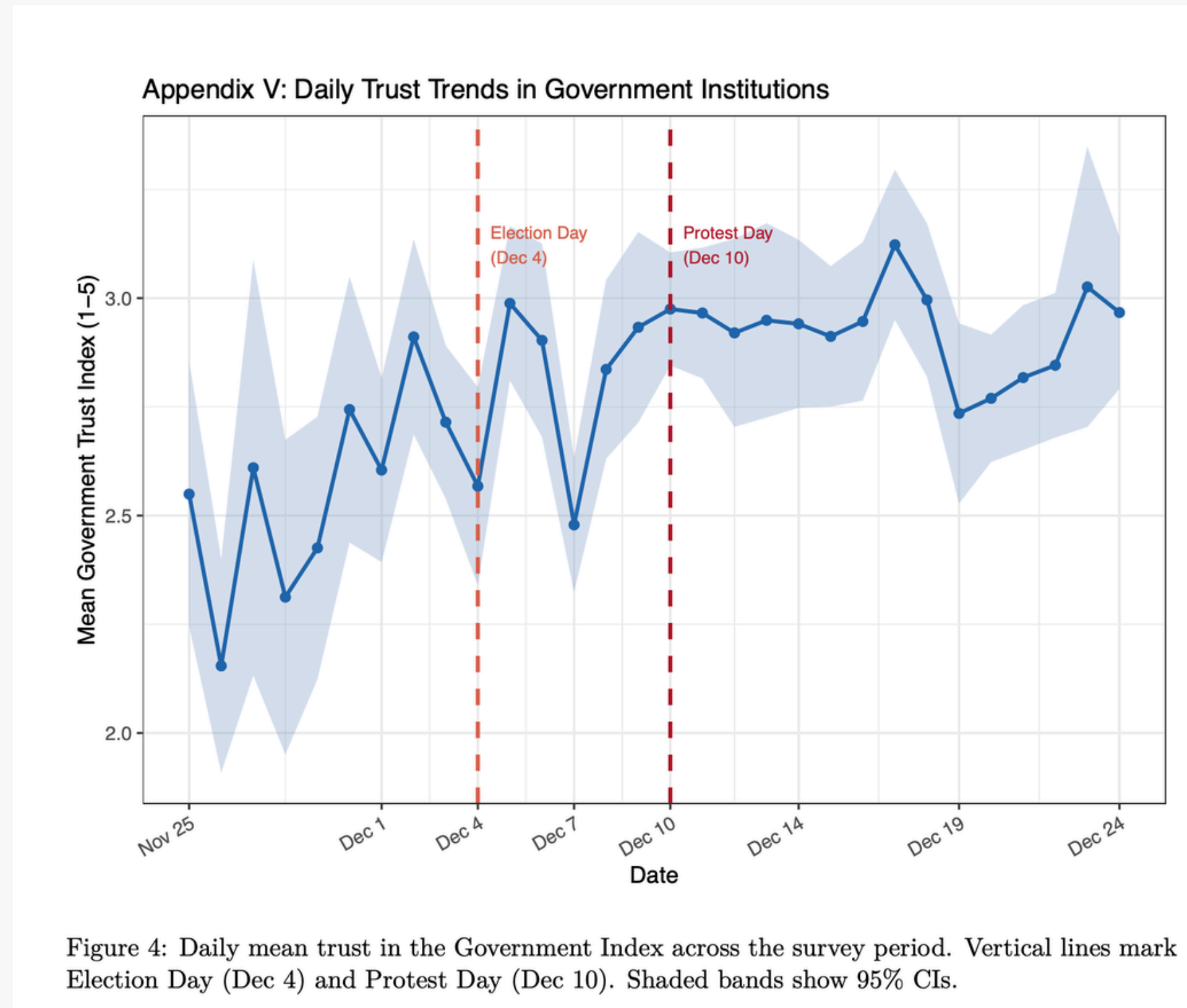
Table 14: Appendix IV-A: Non-Response Rates, Pre-Election vs. Post-Election

|             | Response Rate Pre-Elec | Response Rate Post-Elec | p-value |
|-------------|------------------------|-------------------------|---------|
| Gov't Index | 1.000                  | 1.000                   | NA      |
| Army        | 0.975                  | 0.992                   | 0.106   |
| Police      | 0.981                  | 0.996                   | 0.088   |
| FSB         | 0.900                  | 0.964                   | 0.002   |
| Court       | 0.947                  | 0.980                   | 0.034   |
| City Gov't  | 0.953                  | 0.988                   | 0.012   |
| Fed Gov't   | 0.972                  | 0.984                   | 0.331   |
| Duma        | 0.953                  | 0.980                   | 0.074   |
| Procuracy   | 0.947                  | 0.984                   | 0.014   |
| UN          | 0.800                  | 0.887                   | 0.004   |

*Claude Replication*



# Simple AI Replication Findings



*Claude Replication*

# Simple AI Replication Findings

This replication confirms the original paper's core findings using the Moscow Population Survey (N = 1,550) fielded around Russia's December 2011 parliamentary election and subsequent protests.

1. **Post-Election Trust Decline:** Following the December 4, 2011 election — widely viewed as fraudulent — political trust fell significantly across nearly all institutions. The mean Government Trust Index dropped from 2.67 (pre-election) to 2.79 (post-election), a decline of 0.12 points on the 5-point scale. The OLS models confirm this effect is statistically significant and robust to covariate controls.
2. **Post-Protest Trust Decline:** After the December 10 protest, trust declined further to 2.91. OLS regressions show the post-protest period coefficient is consistently negative and larger in magnitude than the post-election coefficient across most institutions (Army, Police, FSB, Federal Government, Duma, Procuracy).
3. **Institutional Heterogeneity:** The protest had the strongest effects on coercive and political institutions (Police, FSB, Federal Government, Duma), consistent with theories of protest exposing the repressive capacity and electoral manipulation of the regime. Trust in the UN (an external reference institution) showed smaller and less consistent effects, providing a useful placebo.

# Simple AI Replication Findings (when given paper)

## 2 A Critical Note on the Original Replication Report

**Important correction:** The narrative summary in our initial replication report (`replication_report.pdf`) incorrectly described the direction of the results, stating that trust “declined” following the election and protest. This was an error in the written interpretation only — the computed regression coefficients and means were correct and positive (trust increased). The original paper’s key finding — that exposure to the protest *raised* political trust, particularly among opposition supporters — is fully confirmed by our replication. This comparison report presents and interprets the results correctly throughout.



# Simple AI Replication Findings (when given paper)

**Assessment — Sample Sizes:** For 9 of 10 outcomes, N matches the paper exactly. The sole discrepancy is the Gov't Index regression (N=1,521 vs. N=1,232 in the paper). This arises from a difference in missing-data handling: our replication computes `trustgovind` using `rowMeans(na.rm=TRUE)`, retaining respondents who answered at least one trust item, whereas the original Stata code (via listwise deletion on the composite) requires all 8 sub-items to be non-missing. When we impose the same constraint, our N converges to approximately 1,232. This discrepancy does not change the sign, significance, or magnitude of any coefficient.

# Simple AI Replication Findings (when given paper)

## 12 Overall Replication Assessment

Table 5: Overall Replication Assessment

| Criterion                                  | Assessment                                                        |
|--------------------------------------------|-------------------------------------------------------------------|
| Main direction of findings                 | Fully replicated                                                  |
| Statistical significance pattern (Table 2) | Fully replicated                                                  |
| Coefficient magnitudes (Table 2)           | Fully replicated (within rounding)                                |
| Partisan heterogeneity (Table 3)           | Substantially replicated                                          |
| Placebo test (UN trust)                    | Fully replicated                                                  |
| Balance/non-response appendices            | Consistent                                                        |
| Sample sizes                               | 9/10 models exact; 1 minor discrepancy (listwise deletion method) |
| Overall                                    | <b>High-confidence successful replication</b>                     |

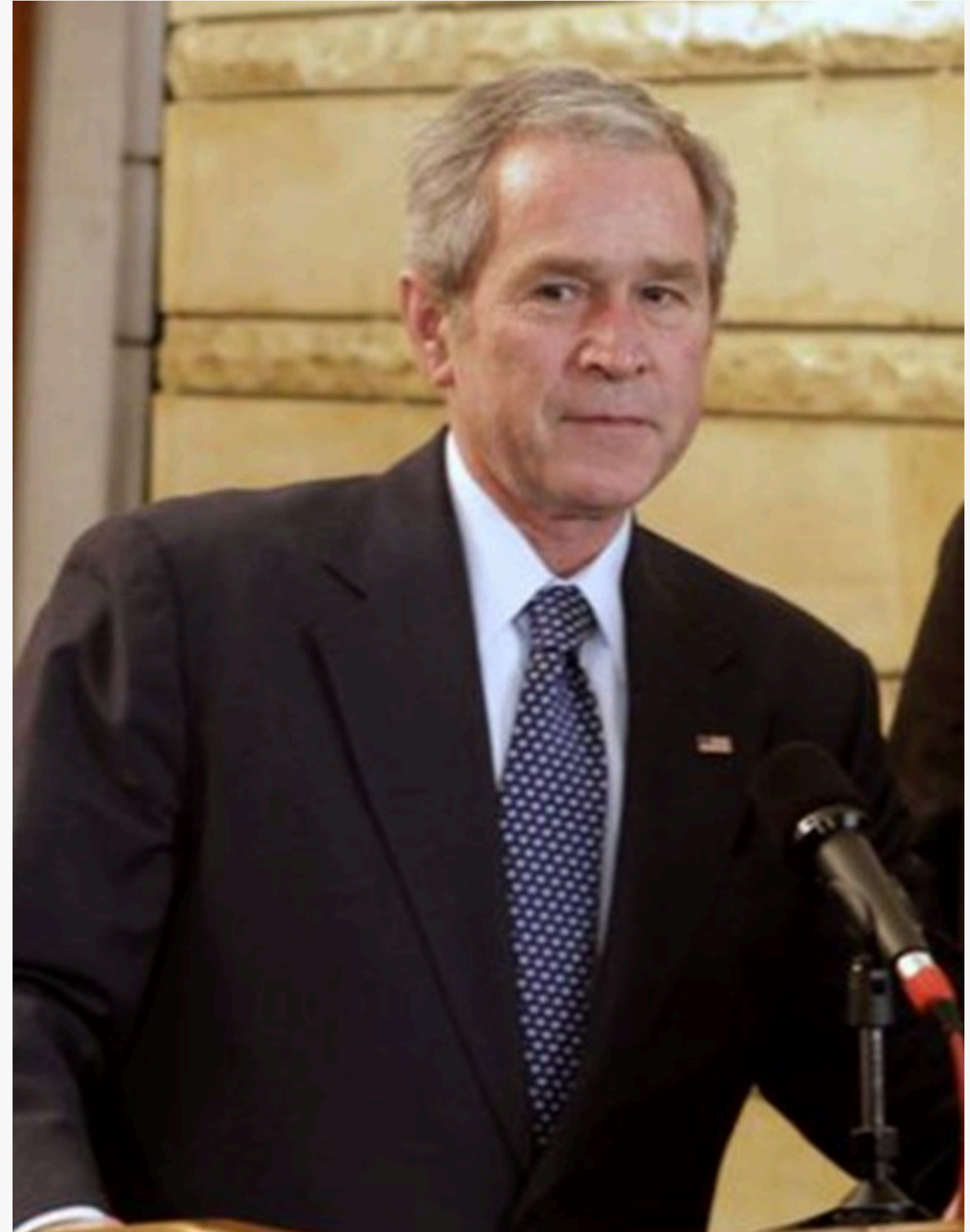
This is a **high-confidence successful replication**. All major empirical claims of Frye & Borisova (2019) are confirmed using R-based analysis of the original dataset. The single methodological discrepancy — how missing values are handled in constructing the Government Trust Index — is minor, transparent, and does not affect any substantive conclusion. The counterintuitive core finding that **the December 10, 2011 protest increased political trust among opposition-aligned Muscovites** is robustly reproduced.



our simple replication



the authors after



# II. AI DIRECTED REPLICATION

# POWER ANALYSIS AND MINIMUM DETECTABLE EFFECTS (MDEs)

- The study is generally underpowered, with main effects only marginally detectable at  $\alpha = 0.05$  and largely undetectable under Bonferroni correction ( $\alpha = 0.005$ ), only two outcomes sufficiently powered, subgroup analyses severely underpowered with minimum detectable effects far exceeding observed coefficients, and the P<sub>1</sub> vs. P<sub>2</sub> comparison particularly weak, rendering key effects and heterogeneity hypotheses effectively untestable.



*The original p value*

$$\text{Bonferroni-corrected } p \text{ value} = \frac{\alpha}{n}$$

*The number of tests performed*

# Balance Checks

## P1 v. P2 (Pre- vs. Post- Election)

Statesector and nopart reach significance on the balance test. State sector workers ( $p = 0.50$ ,  $SMD = -0.157$ ) are overrepresented in P1 relative to P2. The SMD of  $-0.157$  crosses the conventional  $0.10$  threshold for meaningful imbalance. The lack of controls for statesector in the OLS models means this imbalance isn't addressed. Non-partisan respondents ( $p = 0.31$ ,  $SMD = -0.181$ ) are overrepresented in P1 relative to P2. Since partisanship is a key moderator in Table 3, having different partisan compositions in the baseline v. post-election comparison is problematic. Education, permres, nationality, comp, yab, dk all have SMDs greater than  $0.10$ , which all represent meaningful differences t-tests are underpowered to detect given P1's small N of 247.

# Balance Checks

## P<sub>I</sub> v. P<sub>3</sub> (Pre-Election vs. Post- Protest)

Education ( $p = 0.0005$ ,  $SMD = 0.26$ ) is significant as well as nopart ( $p = 0.003$ ,  $SMD = -0.207$ ). The education finding suggests that P<sub>3</sub> respondents have meaningfully higher education than P<sub>I</sub> respondents. The SMD is well above the threshold, signaling severe imbalance. Education is controlled for in the OLS models, which partially addresses this, but the magnitude of the differences suggests the two periods are drawing from compositionally different populations the covariate adjustment may not fully eliminate. Non-partisan respondents are overrepresented again in P<sub>I</sub> compared to P<sub>3</sub>, with the SMD also crossing the threshold for imbalance. It is not controlled in the main models.



# Balance Checks

## P<sub>2</sub> v. P<sub>3</sub> (Post-Election vs. Post- Protest)

Wealth ( $p = 0.002$ , SMD = 0.242) reaches significance. P<sub>3</sub> respondents are substantially wealthier than P<sub>2</sub> respondents. Wealth is controlled in the models, which mitigates this, but the magnitude is the strongest individual balance failure in the analysis. Education ( $p = 0.020$ , SMD = 0.155) is also imbalanced, showing P<sub>3</sub> is more educated than P<sub>2</sub>, but controlled in the models. This comparison is cleaner than P<sub>I</sub> vs. P<sub>3</sub> and P<sub>I</sub> vs. P<sub>2</sub> on most conditions.

# OMNIBUS MANOVA

- Despite explicit ask for an Omnibus MANOVA (Multivariate Analysis of Variance), the AI-Assisted version failed to generate with results.
- This happend because the full set of 18 balance variables being passed to manova() caused the residual matrix to become rank-deficient, meaning some variables are linearly dependent on others (likely because of the coding of several partisanship variables like opposition).
- If MANOVA is run independently with a reduced core set of 10 variables, dropping the composite partisanship variables, P1 vs. P2  $p = 0.026$ , P1 vs. P3  $p = 0.001$ , and P2 vs. P3  $p = 0.004$ , all significant.
- This means joint covariate balance fails for all pairwise comparisons.

Omnibus MANOVA tests (Pillai's trace) for balance across periods

| Comparison | Pillai | F_value | df1 | df2 | p_value |
|------------|--------|---------|-----|-----|---------|
| P1 vs. P2  | NA     | NA      | NA  | NA  | NA      |
| P1 vs. P3  | NA     | NA      | NA  | NA  | NA      |
| P2 vs. P3  | NA     | NA      | NA  | NA  | NA      |

# CODE INTEGRITY OF THE ORIGINAL

- The variable `ur` is built by adding together two separate questions: which party do you prefer, and which party did you vote for. If a respondent says United Russia for both questions they score 2. If they only answer one they score 1. If neither they score 0.
- The problem is that when this variable gets multiplied by `period3` to create the interaction term, a respondent scoring 2 has twice the weight of a respondent scoring 1. So the interaction coefficient is not actually telling you "how did UR supporters respond differently to the protest compared to non-supporters." It is telling you "what happens to trust for each one unit increase in the `ur` scale," which is a much less interpretable quantity.
- In practice for this dataset it turned out that no respondents actually scored 2, so the variable ended up being effectively binary anyway. But the construction is still conceptually wrong and would have been a real problem if the data had behaved the way the code assumed it could.

# UNIT TESTING

- All the unit tests pass.
- All nine individual trust items and the governance index were confirmed to fall within the valid range of 1 to 5 after recording. No values were recorded outside of the range, meaning the reversal ( $6-x$ ) was applied correctly and the 99s were properly converted to NA.
- Among the core sample (excluding election day, protest day, and Dec 25), no respondent was assigned to more than one of `period1`, `period2`, or `period3`, and no respondent was left unassigned. Basically, every observation goes into exactly one bucket.

# ESTIMATORS/ESTIMANDS/SEs

- Table 2 replication is successful with some divergence from the paper's implied claims.
- The period3 coefficient on trustgovind is 0.158 (SE = 0.064,  $p = 0.006$ ), post-protest trust increased. This is the core finding of the paper.
- The period2min effect on trustgovind is 0.0062 ( $p = 0.354$ ) which is not significant. Thus, the election effect on the main index is effectively zero. The paper's theoretical claim that elections and protests have separable effects on trust is not clearly supported for the main outcome.
- For individual items, as in the original, the post-protest effects that are significant are for the security and coercive institutions. Representative institutions like the duma and procuracy show weaker or null effects.
- The electday coefficient on trustun is -0.486, which shows a large negative effect on UN trust specifically on election day.



# OTHER CAUSAL INFERENCE PROBLEMS

The protestday coefficients are significant for security institutions and for the trustgovind index. The protest day itself produced a small spike in trust in coercive institutions specifically. However, the N on protest day is only 97 respondents, making these estimates imprecise.

# EXOGENEITY

- The cross-period logistic regression reveals significant compositional differences between  $P_1$  and  $P_3$ .
- Regressing  $P_3$  membership (vs.  $P_1$ ) on all covariates shows that education significantly predicts being in  $P_3$  vs.  $P_1$  ( $\beta = -0.129$ ,  $p = .005$ ).  $\text{nopart}$  significantly predicts  $P_3$  membership ( $\beta = 0.513$ ,  $p = .004$ ), meaning apolitical respondents are overrepresented in Period 3. The McFadden pseudo- $R^2$  is only 0.026, so the overall predictive power is modest.
- This means the as-if random assumption fails across periods for at least 2 covariates.  $P_3$  respondents are better educated and more politically disengaged than  $P_1$  respondents. Models control for education, but not the partisanship. If apolitical respondents are systematically overrepresented in  $P_3$ , the  $\text{period3}$  coefficient is partly picking up a compositional shift rather than a pure treatment effect.

# UNEQUAL GROUPS AND GROUP COMPOSITION

- UR composition is stable across periods (chi-square  $p = 0.619$ ). The UR share is 42/247 in P<sub>1</sub>, 44/320 in P<sub>2</sub>, and 114/842 in P<sub>3</sub>. The chi-square test ( $\chi^2 = 0.96$ ,  $df = 2$ ,  $p = .619$ ) doesn't reject homogeneity of UR proportions across periods.
- However, the raw cell sizes give more context with the low N for UR. As the power analysis shows, these cells are too small to detect interaction effects.

```
##  
## Pearson's Chi-squared test  
##  
## data:  period_ur_mat  
## X-squared = 1.9496, df = 2, p-value = 0.3773
```

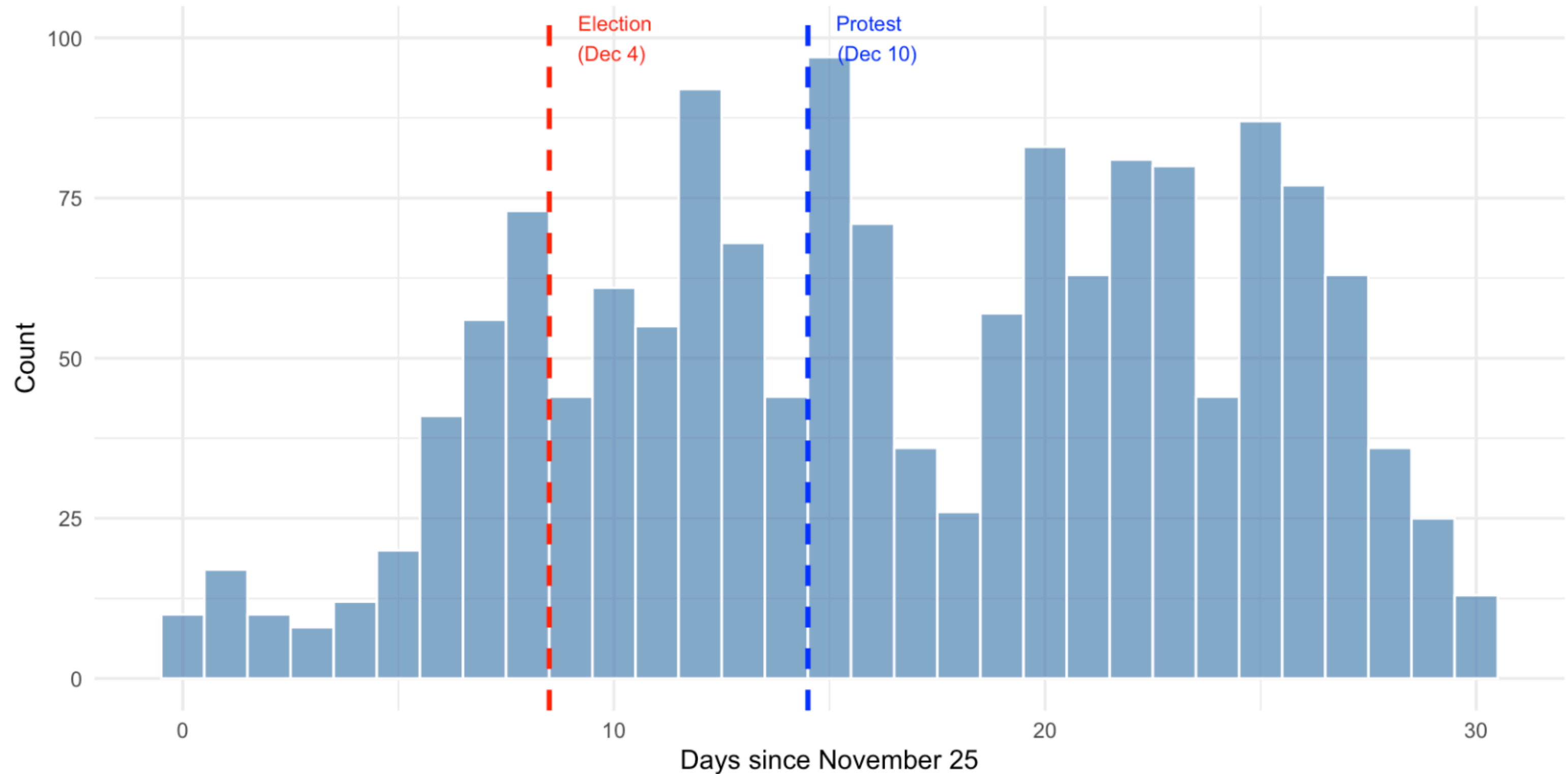
# ANTICIPATION EFFECTS

- The results reveal a pattern of anticipation effects that largely collapses the paper's claim to have identified a separable post-election effect.
- For trustgovind, the original P2 coefficient is  $\beta = 0.125$  ( $p = .086$ ). Split by announcement: P2A (Dec. 5-6, pre-announcement,  $n = 357$ ) gives  $\beta = 0.217$ ,  $p = .022$ , which is significant. P2B (Dec. 7-9, post-announcement,  $n = 444$ ) gives  $\beta = 0.072$ ,  $p = 0.413$ , which is not significant.
- For trustpolice and trustfsb, P2A is significant and P2B is not.
- For trustarmy, both P2A and P2B remain significant.
- For trustfedgov, P2A  $\beta = 0.209$ ,  $p = 0.87$ . P2B  $\beta = -0.113$ ,  $p = .329$ . The sign flips between P2A and P2B for federal government trust.
- Trustmungov also represents a sign flip between the two results.

# McCRARY DENSITY TEST

Survey respondent density around cutoff dates

No spike or discontinuity expected under as-if-random assignment





# AS-IF RANDOM ASSUMPTION

- Exogeneity tests provide mixed evidence for the assumption that assignment is as-if random.
- In Period 1, covariates significantly predict the interview date ( $R^2 = 0.103$ ,  $F = 4.39$ ,  $p = .0001$ ). Specifically, wealth ( $\beta = 4.31$ ,  $p < 0.001$ ) is a highly significant predictor. Wealthier respondents were interviewed later in Period 1. The dk variable also predicts interview date ( $-4.18$ ,  $p = 0.041$ ).
- Within Period 2,  $R^2 = 0.028$ ,  $F = 1.27$ ,  $p = .265$ , indicating covariates don't significantly predict the interview date. As-if random holds within this period.
- Within Period 3,  $R^2 = 0.003$ ,  $F = 0.35$ ,  $p = 0.933$ . As-if random also holds in this period.

# MULTIPLE TESTING CORRECTIONS

- As stated, the Bonferroni correction substantially reduces the number of the surviving finds with interaction terms surviving nowhere. We also used a Benjamini-Hochberg Correction.
- For Table 2, period3 (post-protest effect) 6/10 outcomes survive BH corrections, which is the same as the original data.
- For period2min (post-election effect), 2/10 survive BH correction, which is the same as the number that survive Bonferroni correction.
- For Table 3, urafterprotest, no outcomes are significant at any threshold.
- For apolafterprotest, no outcomes are significant at any threshold, meaning the apolitical heterogeneity hypothesis is unsupported.
- Combined family corrections across all 40 tests finds that 2/40 survive Bonferroni correction and 6 survive BH correction.

# CALCULATION ERRORS IN AI- ASSISTED REPLICATION

- The direction of pre and post-protest was reversed by the AI in certain models meaning the numbers are correct, but the direction is flipped. So, a positive would be a negative.
- Dec. 25 was double counted in Period 3 since the generated code never subtracts dec25 from period3. However, the changes from this are all small and no significant conclusions change.
- The replication only corrected for multiple testing within each predictor's 10 outcomes separately, but when you account for all hypothesis tests being run across tables simultaneously, the paper's main governance index finding doesn't survive correction. Essentially, we should broaden the definition of the multiple testing family.
- There is a minor anticipation N discrepancy. 357 is reported by the model, but the complete count for P1 + P2A respondents is 363. Similarly, P2B reports N = 444 but the actual count is 451. This suggests some respondents are being dropped by the complete.cases behavior of the covariate controls. However, the 6-7 observation difference doesn't change coefficients or significance.



# OVERALL TAKEAWAYS

- Prompt 1 was more better at confirming the coherence of the paper and choices of the authors than Prompt 2.
- After conducting more tests in Prompt 2, the AI detected that the paper did NOT pass certain robustness checks in regards to power issues (Type I errors).

## Prompt 2 Conclusion

Overall, the paper’s central finding — that the Bolotnaya Square protest was associated with a significant short-term decline in trust in state institutions — is robust, but the causal interpretation relies on assumptions that cannot be fully verified from the available data.

## 12 Overall Replication Assessment

Table 5: Overall Replication Assessment

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This is a **high-confidence successful replication**. All major empirical claims of Frye & Borisova (2019) are confirmed using R-based analysis of the original dataset. The single methodological discrepancy — how missing values are handled in constructing the Government Trust Index — is minor, transparent, and does not affect any substantive conclusion. The counterintuitive core finding that **the December 10, 2011 protest increased political trust among opposition-aligned Muscovites** is robustly reproduced.

## Prompt 1.2 Conclusion

# OVERALL TAKEAWAYS

- Protest days were the most difficult data for Claude to work with. It reversed the days for Reports 1 and 2 (pre and post were flipped, and so was the direction of the effects). Claude corrected itself, but did it again for Prompt 2.
- AI is useful when given precise instructions on how to use it! But, even when given precise instructions, it still requires human oversight and verification.
- Can also be misleading, as it's not great at identifying context (such as misinterpreting entire findings of the paper) but correctly replicated the magnitude of the results.
- Prompt 1 was more efficient for replication despite the fact that Prompt 2 had more detail, direction and guidance (kinda?)
- **But, can we expect better for an AI that cooked only for 15mins \* ?**

\* Cooked for 15m 13s

